

# Mathematical Logic

沈威宇

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# 1 Mathematical Logic

## I Proof Theory

### i Deductive system, proof system, formal system, formal proof system, proof calculus, or deductive calculus

A deductive system, proof system, formal system, or formal proof system  $\Gamma$  over a formal language  $\mathcal{L}$  consists of axioms and rules of inference, where:

- An axiom is a formula in  $\mathcal{L}$  assumed to be derived in  $\Gamma$ .
- An axiom schema is an expression with schema variables, each designated with a type of symbols in  $\mathcal{L}$ , that the expression obtained by replacing each schema variable with an arbitrary symbol of its type, called an application of the axiom schema, is assumed to be an axiom.
- A rule of inference in Hilbert style is in the form  $\frac{P_1, P_2, \dots, P_n}{Q}$  where  $P_1, P_2, \dots, P_n, Q$  are expressions sharing the same set of metaformulas such that any expressions obtained by replacing each metaformula with an arbitrary formula in  $\mathcal{L}$  are formulas in  $\mathcal{L}$  and the expressions obtained by such replacement of  $P_1, P_2, \dots, P_n$  are assumed to derive the expression obtained by such replacement of  $Q$ , in which  $P_1, P_2, \dots, P_n$  and their replacements are called premisses and  $Q$  and their replacements called conclusion. Other styles exist and have the same expressive power as Hilbert style for propositional logic and predicate logic.

### ii (Formal) derivation or proof

Given proof system  $\Gamma$  over language  $\mathcal{L}$  and formulas  $S_1, S_2, \dots, S_n, T \in \mathcal{L}$ , a (formal) derivation or proof from  $S_1, S_2, \dots, S_n$  to  $T$  is a sequence of formulas  $\langle F_i \rangle_{i=1}^m$  such that each  $F_i$  is either

- an axiom of  $\Gamma$ ,
- an element of  $\{S_1, S_2, \dots, S_n\}$ , or
- can be derived from some of the earlier formulas in the sequence using a rule of inference,

and that  $F_m = T$ .

$S_1, S_2, \dots, S_n$  are called premisses,  $T$  is called conclusion, and this process is called argumentation.

That there exists a derivation from  $S_1, S_2, \dots, S_n$  to  $T$  is denoted as  $S_1, S_2, \dots, S_n \vdash T$  and called  $S_1, S_2, \dots, S_n$  derive or prove  $T$ .

Thus, for any axiom  $\alpha$ ,  $\vdash \alpha$ .

Any formula  $\alpha$  such that  $\vdash \alpha$  is called derived or a theorem and is also denoted as  $\Gamma \vdash \alpha$ .

### iii Finitarity

A rule of inference is finitary iff its premises are finite.

The rules of inference and sequences in formal proofs are required to be finite for finitary deductive systems.

### iv Soundness

A deductive system is sound if every formula that can be derived from the system is a tautology in the language.

### v Syntactic consequence

For formulas  $S_1, S_2, \dots, S_n, T$  in a language  $\mathcal{L}$ ,  $T$  is a syntactic consequence of  $S_1, S_2, \dots, S_n$ , denoted as  $S_1, S_2, \dots, S_n \vdash T$ , iff there exists a sound proof system over  $\mathcal{L}$  in which  $S_1, S_2, \dots, S_n \vdash T$ .

### vi Independent formula

A formula  $P$  such that neither  $\vdash P$  nor  $\vdash \neg P$ .

### vii (Syntactical) completeness

A deductive system is (syntactically) complete if for every formula in the language, either it or its negation can be derived from the system.

### viii Semantical completeness

A deductive system is semantically complete if every tautology in the language can be derived from the system.

### ix Interpretability

If a deductive system  $S$  over language  $L$  and a deductive system  $T$  over language  $M$  are such that there exists a function  $H: L \rightarrow M$  such that  $S \vdash l \implies T \vdash H(l)$ , then we say  $T$  interprets  $S$ ,  $S$  is weaker than  $T$ , or  $T$  is stronger than  $S$ .

### x Computably enumerable (c.e.), recursively enumerable (r.e.), semidecidable, partially decidable, listable, provable or Turing-recognizable deductive system

A deductive system is computably enumerable (c.e.), recursively enumerable (r.e.), semidecidable, partially decidable, listable, provable or Turing-recognizable iff the set of all axioms of it is computably enumerable.

If a deductive system is computably enumerable and its rules of inferences are finitary, it is finitary.

## II Model Theory

### i Signature

A signature  $\sigma = (\mathcal{P}, \mathcal{F})$  is a pair of indexed sets,

- a set of sets of predicate symbols (also called relation symbols)  $\mathcal{P} = \{P_n \mid n \in \mathbb{N}_0\}$  with  $P_n$  being a possibly infinite set of predicate symbols of arity  $n$ ,
- a set of sets of function symbols  $\mathcal{F} = \{F_n \mid n \in \mathbb{N}_0\}$  with  $F_n$  being a possibly infinite set of function symbols of arity  $n$ , with any element of  $F_0$  called a constant.

### ii (Logical or model-theoretic) structure

A (logical or model-theoretic) structure can be defined with a triple  $(A, \sigma, I)$  of a set called domain, universe, carrier (set), underlying set, domain of discourse, or universe of discourse,  $A$ , denoted as  $|\mathcal{A}|$  or simply  $\mathcal{A}$ , a signature  $\sigma$ , and an interpretation function  $I$  that, maps for all  $n \in \mathbb{N}_0$ , maps each  $f \in F_n$  to an  $n$ -ary operation  $I(f)$  on  $A$ , called interpretation of  $f$  and often denoted as  $f$ , and for all  $n \in \mathbb{N}_0$ , maps each  $p \in P_n$  to a subset  $I(p)$  of  $A^n$ , called interpretation of  $p$  and often denoted as  $p$ , in which  $A^0 = \{\emptyset\}$ .

A structure with signature  $\sigma$  is called a  $\sigma$ -structure. A structure with signature  $\sigma$  of a language  $L$  is called a  $L$ -structure.

### iii Substructure

A structure  $\mathcal{B} = (B, \sigma_B, I_B)$  is a substructure of a structure  $\mathcal{A} = (A, \sigma_A, I_A)$ , that is,  $\mathcal{A}$  is an extension of  $\mathcal{B}$ , denoted as  $\mathcal{B} \subseteq \mathcal{A}$ , iff

- $B \subseteq A$ ,
- $\sigma_B = \sigma_A$ ,
- for every  $n \in \mathbb{N}_0$ , for every function symbol  $f \in F_n$ ,  $I_B(f) = I_A(f) \upharpoonright B^n$ , and
- for every  $n \in \mathbb{N}_0$ , for every relation symbol  $p \in P_n$ ,  $I_B(p) = I_A(p) \cap B^n$ .

### iv Elementary substructure

A structure  $\mathcal{B} = (B, \sigma_B, I_B)$  is an elementary substructure of a structure  $\mathcal{A} = (A, \sigma_A, I_A)$ , that is,  $\mathcal{A}$  is an elementary extension of  $\mathcal{B}$ , denoted as  $\mathcal{B} < \mathcal{A}$ , iff, with the language of  $\sigma_B$  denoted as  $\mathcal{L}$ ,

- $B \subseteq A$ ,
- for every  $n$ -ary formula  $\varphi \in \mathcal{L}$ , for every  $a \in B^n$ ,  $\mathcal{B} \models \varphi[a] \iff \mathcal{A} \models \varphi[a]$ .

### v Closed subset

For a structure  $\mathcal{A} = (A, \sigma, I)$ , a subset  $B \subseteq A$  is called closed iff, for all  $n \in \mathbb{N}_0$ , for any  $f \in F_n$  and  $b \in B^n$ ,  $I(f)(b) \in B$ .

For any subset  $B$  of  $A$ , there exists a smallest closed subset that is a superset of  $B$ , called the hull of  $B$  or the closed subset derived by  $B$ , denoted as  $\langle B \rangle_A$ .

## vi Cardinality

The cardinality of a structure is defined as the cardinality of its domain.

## vii Homomorphism

A  $(\sigma-)$ homomorphism from a structure  $\mathcal{A} = (A, \sigma, I_A)$  to a structure  $\mathcal{B} = (B, \sigma, I_B)$  is a map  $h: A \rightarrow B$  such that,

- for all  $n \in \mathbb{N}_0$ , for any  $f \in F_n$  and  $a_1, a_2, \dots, a_n \in A^n$ ,

$$h(I_A(f)(a_1, a_2, \dots, a_n)) = I_B(f)(h(a_1), h(a_2), \dots, h(a_n)),$$

- for all  $n \in \mathbb{N}_0$ , for any  $p \in P_n$  and  $a_1, a_2, \dots, a_n \in A^n$ ,

$$(a_1, a_2, \dots, a_n) \in I_A(p) \implies (h(a_1), h(a_2), \dots, h(a_n)) \in I_B(p).$$

For any  $a \in A^n$  with  $a = (a_1, a_2, \dots, a_n)$ ,  $(h(a_1), h(a_2), \dots, h(a_n))$  is denoted as  $h(a)$ .

## viii Strong homomorphism

A strong  $(\sigma-)$ homomorphism from a structure  $\mathcal{A} = (A, \sigma, I_A)$  to a structure  $\mathcal{B} = (B, \sigma, I_B)$  is a map  $h: A \rightarrow B$  such that,

- for all  $n \in \mathbb{N}_0$ , for any  $f \in F_n$  and  $a_1, a_2, \dots, a_n \in A^n$ ,

$$h(I_A(f)(a_1, a_2, \dots, a_n)) = I_B(f)(h(a_1), h(a_2), \dots, h(a_n)),$$

- for all  $n \in \mathbb{N}_0$ , for any  $p \in P_n$  and  $a_1, a_2, \dots, a_n \in A^n$ ,

$$(a_1, a_2, \dots, a_n) \in I_A(p) \iff (h(a_1), h(a_2), \dots, h(a_n)) \in I_B(p).$$

## ix Embedding

A  $(\sigma-)$ embedding from a structure  $\mathcal{A} = (A, \sigma, I_A)$  to a structure  $\mathcal{B} = (B, \sigma, I_B)$  is a strong homomorphism  $h$  from  $\mathcal{A}$  to  $\mathcal{B}$  that is injective.

## x Isomorphism

A  $(\sigma-)$ isomorphism from a structure  $\mathcal{A} = (A, \sigma, I_A)$  to a structure  $\mathcal{B} = (B, \sigma, I_B)$  is a strong homomorphism  $h$  from  $\mathcal{A}$  to  $\mathcal{B}$  that is bijective.

## xi Isomorphism-invariance

An  $n$ -free variable formula  $\varphi$  is isomorphism-invariant between two models  $\mathcal{M}, \mathcal{N}$  with domains  $M, N$  iff for every isomorphism  $h$  between  $\mathcal{M}, \mathcal{N}$ , for any  $a \in M^n$ :

$$\mathcal{M} \vdash \varphi[a] \iff \mathcal{N} \vdash \varphi[h(a)].$$



## xii Definability

Let  $L$  be a first-order language,  $\mathcal{M}$  be a  $L$ -structure with domain  $M$ ,  $X$  be a fixed subset of  $M$ , and  $m$  be a positive integer.

- A set  $A \subseteq M^m$  is definable in  $\mathcal{M}$  with parameters from  $X$  iff there exists a formula  $\varphi$  of  $m + n$  free variables and  $b_1, b_2, \dots, b_n \in X$  such that  $\forall a_1 \in M \forall a_2 \in M \dots \forall a_m \in M, (a_1, a_2, \dots, a_m) \in A$  iff  $\mathcal{M} \models \varphi[a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n]$ , and  $\varphi[a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n]$  with parameters  $b_1, b_2, \dots, b_n$  is called defining formula.
- A set is definable in  $\mathcal{M}$  without parameter iff it is definable in  $\mathcal{M}$  with parameters from  $\emptyset$ .
- A function is definable in  $\mathcal{M}$  (with some parameters) iff its graph is definable in  $\mathcal{M}$  (with those parameters).
- An element  $a \in M^m$  is definable in  $\mathcal{M}$  (with some parameters) iff  $\{a\}$  is definable in  $\mathcal{M}$  (with those parameters).
- Set of definable sets:  $\text{Def}(\mathcal{M})$  is the union for all  $m \in \mathbb{N}$  of the set of subsets of  $M^m$  that is definable in  $\mathcal{M}$  (with any parameter or without parameter).

## III Propositional logic (PL), statement logic, sentential logic, propositional calculus, statement calculus, sentential calculus, or zeroth-order logic

### i Syntax

Given

- a set of connectives, logical connectives, logical operators, truth-functional connectives, truth-functors, or propositional connectives  $\mathcal{O} = \{\neg, \wedge, \vee\}$  with unary negation  $\neg$ , binary conjunction  $\wedge$ , and binary disjunction  $\vee$ , (or any functionally complete set of logical operators in Boolean algebra with related definitions changed accordingly), and
- a set of atomic propositions, atomic formulas, atomic sentences, atoms, proposition letters, sentence letters, propositional variables, or variables  $\mathcal{P}$  with  $\mathcal{P} \cap \mathcal{O} = \emptyset$  and  $\mathcal{O} \cup \mathcal{P}$  be the alphabet of  $\mathcal{L}$ ,

the formal language  $\mathcal{L}$  of propositional logic is defined by:

- $\mathcal{P} \subseteq \mathcal{L}$ ,
- $\neg P \in \mathcal{L}$  if  $P \in \mathcal{L}$ ,
- $(P \wedge Q) \in \mathcal{L}$  if  $P, Q \in \mathcal{L}$ ,
- $(P \vee Q) \in \mathcal{L}$  if  $P, Q \in \mathcal{L}$ , and
- nothing else is in  $\mathcal{L}$ .

Any  $w \in \mathcal{L}$  is called a well-formed formula (WFF or wff), or simply formula.

## ii Semantics

For a given propositional logic, its interpretation (or valuation) is defined as an interpretation (or valuation) function of its set of atomic propositions.

For a given set of atomic propositions  $\mathcal{P}$ , an interpretation (or valuation) function  $I$  maps every formula in  $\mathcal{P}$  to a semantic value in  $\{1, 0\}$  (with 1 meaning true and 0 meaning false), denoted, for each  $P \in \mathcal{P}$ , as

$$I(P) = B.$$

For a given propositional logic with formal language  $\mathcal{L}$  and interpretation  $I$ , the interpretation function  $\mathcal{J}$  over  $\mathcal{L}$  is recursively defined as

- for any  $P \in \mathcal{P}$ ,  $\mathcal{J}(P) = I(P)$ ,
- for any  $P \in \mathcal{L}$ ,  $\mathcal{J}(\neg P) = \neg \mathcal{J}(P)$ ,
- for any  $P, Q \in \mathcal{L}$ ,  $\mathcal{J}(P \wedge Q) = \mathcal{J}(P) \wedge \mathcal{J}(Q)$ , and
- for any  $P, Q \in \mathcal{L}$ ,  $\mathcal{J}(P \vee Q) = \mathcal{J}(P) \vee \mathcal{J}(Q)$ .

For any  $P \in \mathcal{L}$ , we write  $I \models P$  if  $\mathcal{J}(P) = 1$  and  $I \not\models P$  if  $\mathcal{J}(P) = 0$ .

Alternatively, with structure, as expressed in the language of first-order logic, we can define each atomic formula in propositional logic as a nullary predicate symbol, that there is no function symbol, and that the interpretation structure maps each nullary predicate symbol to a subset of  $\{\emptyset\}$ , that is, either  $\{\emptyset\}$  or  $\emptyset$ , and with  $\emptyset \in \{\emptyset\}$  and  $\emptyset \notin \emptyset$ , an atomic formula mapped to  $\{\emptyset\}$  is true and an atomic formula mapped to  $\emptyset$  is false.

## iii Semantic consequence

Given formulas  $S_1, S_2, \dots, S_n, T$ ,  $S_1, S_2, \dots, S_n \models T$ , called that  $T$  is a semantic consequence of  $S_1, S_2, \dots, S_n$ , means that for any interpretation function  $I$ , if  $I \models S_1, I \models S_2, \dots, I \models S_n$ , then  $I \models T$ .

## iv Equivalence of syntactic and semantic consequences

For any formulas  $S_1, S_2, \dots, S_n, T$ ,

$$S_1, S_2, \dots, S_n \vdash T \iff S_1, S_2, \dots, S_n \models T.$$

Thus syntactic and semantic consequences are called (logical) consequences.

## v Classification of formulas

A formula  $P$  is called (logically) valid or a tautology, denoted as  $\top$ , iff for any interpretation function  $I$ ,  $I \models P$ .

A formula  $P$  is called (logically) invalid or a contradiction, denoted as  $\perp$ , iff for any interpretation function  $I$ ,  $I \not\models P$ .

A formula  $P$  is called contingent iff it is neither a tautology nor a contradiction.

A formula  $P$  is called satisfiable iff it is not a contradiction.

## vi Typical rules of inference

For all equations that hold in Boolean algebra, with each Boolean variable replaced with a metaformula, LHS as numerator and RHS as denominator.

## IV First-order logic (FOL), predicate logic, predicate calculus, or quantificational logic

Proportional logic and predicate logic are collectively called classical logic.

### i Syntax

Given logic symbols,

- (either one of) the quantifier symbols  $\forall$  for universal quantification and  $\exists$  for existential quantification,
- a set of logical connectives  $\mathcal{O} = \{\neg, \wedge, \vee, \implies\}$  with unary negation  $\neg$ , binary conjunction  $\wedge$ , binary disjunction  $\vee$ , and binary material implication  $\implies$ , (or any functionally complete set of logical operators in Boolean algebra with related definitions changed accordingly),
- a set of punctuation symbols  $\mathcal{N}$  containing  $(, ), ,$  and maybe other symbols, which do not carry meaning themselves but to help parse and read the language,
- an infinite set of variables  $\mathcal{V}$ ,
- an equality symbol  $=$  for first-order logic with equality,

and a signature  $\sigma = (\mathcal{P}, \mathcal{F})$  with the equality symbol  $= \in P_2$  for first-order logic with equality, the set of terms  $\mathcal{T}$  is defined by:

- $\mathcal{V} \subseteq \mathcal{T}$ , and
- for all  $n \in \mathbb{N}_0$ , for all  $t \in \mathcal{T}^n$ , for all  $f \in F_n$ ,  $f(t) \in \mathcal{T}$ ,

the set of atomic formulas  $\mathcal{C}$  is defined as

$$\mathcal{C} = \bigcup_{n \in \mathbb{N}_0} \{p(t) \mid t \in \mathcal{T}^n \wedge p \in P_n\},$$

and the formal language  $\mathcal{L}$  of first-order logic is defined by:

- $\mathcal{C} \subseteq \mathcal{L}$
- $\neg P \in \mathcal{L}$  if  $P \in \mathcal{L}$ ,
- $(P \wedge Q) \in \mathcal{L}$  if  $P, Q \in \mathcal{L}$ ,
- $(P \vee Q) \in \mathcal{L}$  if  $P, Q \in \mathcal{L}$ ,
- $(P \implies Q) \in \mathcal{L}$  if  $P, Q \in \mathcal{L}$ , and
- $\forall x P, \exists x P \in \mathcal{L}$  if  $P \in \mathcal{L}$  and  $x \in \mathcal{V}$ ,

- nothing else is in  $\mathcal{L}$ .

Any  $w \in \mathcal{L}$  is called a well-formed formula (WFF or wff) or simply formula.

A variable in a formula is either free or bounded, which are defined as follows for formulas  $P$  and  $Q$ :

- A variable in an atomic formula is free.
- A variable is free in  $\neg P$  if it is free in formula  $P$ .
- A variable is free in  $P \wedge Q$  if it is free in formula  $P$  or  $Q$ .
- A variable is free in  $P \vee Q$  if it is free in formula  $P$  or  $Q$ .
- A variable is free in  $P \implies Q$  if it is free in formula  $P$  or  $Q$ .
- A variable  $x$  is free in  $\forall y P$  and  $\exists y P$  if  $x$  is not the same symbol as  $y$  and  $x$  is free in formula  $P$ .
- No other variable in a formula is free.

A formula with no free variable is called a sentence or a closed formula.

A formula  $P$  with free variables  $x_1, x_2, \dots, x_n$  is denoted as  $P[x_1, x_2, \dots, x_n]$  or  $P(x_1, x_2, \dots, x_n)$ .

The cardinalities of  $\sigma$  and  $\mathcal{L}$  are both defined as the cardinality of  $\left(\bigcup_{u \in \mathcal{P}} u\right) \cup \left(\bigcup_{u \in \mathcal{F}} u\right)$ .

## ii Semantics

For a given first-order logic, its interpretation (or valuation) is defined as  $\mathcal{I} = \mathcal{A}, s$  with an interpretation (or valuation) structure  $\mathcal{A} = (A, \sigma, I)$  of the same signature as the first-order logic with the constraint for first-order logic with equality that

$$I(=) = \{(a, a) \mid a \in A\}$$

and a variable assignment function  $s$  from the variables  $\mathcal{V}$  to  $A$ .

For any formula  $P$ ,  $\mathcal{A}, s \models P$  denotes that  $P$  is true in  $\mathcal{A}$  under assignment  $s$ , and  $\mathcal{A}, s \not\models P$ , denotes that  $P$  is false in  $\mathcal{A}$  under assignment  $s$ .

For a given first-order logic with formal language  $\mathcal{L}$  and interpretation  $\mathcal{A}, s$ ,

- for any term  $t$ ,  $[t]$  or  $[t]_{\mathcal{A}, s}$  denotes  $t$  with any function symbols  $f$  replaced with  $I(f)$  and any variable  $x$  replaced with  $s(x)$ ,
- for any atomic formula  $p(t_1, t_2, \dots, t_n)$  for some predicate symbol  $p \in P_n$ ,  $\mathcal{A}, s \models p(t)$  iff  $([t_1], [t_2], \dots, [t_n]) \in I(p)$ , and
- T-schema or truth schema: with  $\mathcal{J}$  mapping formula  $P$  to 1 if  $\mathcal{A}, s \models P$  and to 0 if  $\mathcal{A}, s \not\models P$ ,
  - for any  $P \in \mathcal{L}$ ,  $\mathcal{J}(\neg P) = \neg \mathcal{J}(P)$ ,
  - for any  $P, Q \in \mathcal{L}$ ,  $\mathcal{J}(P \wedge Q) = \mathcal{J}(P) \wedge \mathcal{J}(Q)$ ,

- for any  $P, Q \in \mathcal{L}$ ,  $\mathcal{J}(P \vee Q) = \mathcal{J}(P) \vee \mathcal{J}(Q)$ ,
- for any  $P, Q \in \mathcal{L}$ ,  $\mathcal{J}(P \implies Q) = \mathcal{J}(P) \implies \mathcal{J}(Q)$ ,
- $\mathcal{A}, s \models \forall x P$  iff  $\forall a \in A$ , replacing  $x$  with  $a$  makes  $\mathcal{A}, s \models P$ , that is,  $\forall x P = \bigwedge_{a \in A} P[x \mapsto a]$ ,  
and
- $\mathcal{A}, s \models \exists x P$  iff  $\exists a \in A$ , replacing  $x$  with  $a$  makes  $\mathcal{A}, s \models P$ , that is,  $\exists x P = \bigvee_{a \in A} P[x \mapsto a]$ .

For sentences,  $\mathcal{A}, s$  can be denoted as  $\mathcal{A}$  since there is no free variable and thus  $s$  is irrelevant.

### iii Semantic consequence

Given formulas  $S_1, S_2, \dots, S_n, T$ ,  $S_1, S_2, \dots, S_n \models T$ , called that  $T$  is a semantic consequence of  $S_1, S_2, \dots, S_n$ , means that for any interpretation  $\mathcal{A}, s$ , if  $\mathcal{A}, s \models S_1, \mathcal{A}, s \models S_2, \dots, \mathcal{A}, s \models S_n$ , then  $\mathcal{A}, s \models T$ .

### iv Equivalence of syntactic and semantic consequences

For any formulas  $S_1, S_2, \dots, S_n, T$ ,

$$S_1, S_2, \dots, S_n \vdash T \iff S_1, S_2, \dots, S_n \models T.$$

Thus syntactic and semantic consequences are called (logical) consequences.

The  $\Leftarrow$  direction is the Gödel's completeness theorem and the  $\Rightarrow$  direction is from the Henkin's model existence theorem.

### v Classification of formulas

A formula  $P$  is called (logically) valid or a tautology, denoted as  $\top$ , iff for any interpretation  $\mathcal{A}, s$ ,  $\mathcal{A}, s \models P$ .

A formula  $P$  is called (logically) invalid or a contradiction, denoted as  $\perp$ , iff for any interpretation  $\mathcal{A}, s$ ,  $\mathcal{A}, s \not\models P$ .

A formula  $P$  is called contingent iff it is neither a tautology nor a contradiction.

A formula is satisfiable iff it is not a contradiction.

### vi Typical rules of inference

- For all equations that hold in Boolean algebra, with each Boolean variable replaced with a metaformula, LHS as numerator and RHS as denominator,
- for any formula  $P$  with one free variable, for any variable  $a$ ,

$$\frac{\forall x P(x)}{P(a)}$$

(universal instantiation), and

- for any formula  $P$  with one free variable, for any variable  $a$ ,

$$\frac{P(a)}{\exists x P(x)}$$

(existential generalization).

## V (Formal) (Deductive) Theory

### i (Formal) (deductive) theory

A (formal) (deductive) theory  $T$  over a formal language  $\mathcal{L}$  is a set of formulas in  $\mathcal{L}$  that is closed under  $\vdash$ . Each  $P \in T$  is called a theorem.

### ii Deductive system

For a theory  $T$  over language  $\mathcal{L}$ , a deductive system that  $T = \{P \in \mathcal{L} \mid \vdash P\}$  is called to derive  $T$  and is not distinguished with  $T$  in terms of properties.

### iii (Syntactical) consistency

A theory  $T$  is (syntactically) consistent iff there is no formula  $P$  such that  $P \in T$  and  $\neg P \in T$ .

### iv Satisfiability

A theory  $T$  is satisfiable iff there exists an interpretation  $\mathcal{I}$  such that for every  $P \in T$ ,  $\mathcal{I} \models P$ , and  $T$  is said to be satisfied by  $\mathcal{I}$  or has an interpretation  $\mathcal{I}$ .

### v Complete (consistent) theory

A complete (consistent) theory  $T$  is a consistent theory such that for every formula  $P$  in its language, either  $T \vdash P$  or  $T \cup \{P\}$  is inconsistent.

### vi Subtheory

If a theory  $S$  and a theory  $T$  is such that  $S \subseteq T$ , then  $S$  is called a subtheory of  $T$ ,  $T$  is called a supertheory or an extension of  $S$ .

### vii Effectively axiomatizable, computably axiomatizable, or recursively axiomatizable

A theory  $T$  over a language  $\mathcal{L}$  is effectively axiomatizable, computably axiomatizable, or recursively axiomatizable if there exists a deductive system with recursively enumerable set of axioms such that  $T = \{P \in \mathcal{L} \mid \vdash P\}$ , that is, there exists a finitary deductive system such that  $T = \{P \in \mathcal{L} \mid \vdash P\}$  and there exists a recursively enumerable set of formal proofs that derive all  $P \in T$ .

### viii Independent formula

For any independent formula  $P$  of a sound deductive system that generate a satisfiable theory  $T$ ,  $T \cup \{P\}$  and  $T \cup \{\neg P\}$  are both satisfiable.

## ix (Complete) theory of interpretation

The (complete) theory of an interpretation  $\mathcal{A}$  is all formulas that are satisfied by  $\mathcal{A}$ , denoted as  $\text{Th}(\mathcal{A})$ .

The theory of a class  $K$  of interpretations is all formulas that are satisfied by all  $k \in K$ , denoted as  $\text{Th}(K)$ .

## VI First-order theory

### i First-order theory

A (formal) (deductive) first-order theory  $T$  over a formal language  $\mathcal{L}$  is a set of sentences in  $\mathcal{L}$  that is closed under  $\vdash$ .

### ii First-order theory with identity

A first-order theory is with identity if its language has the equality symbol  $=$ , and the set of theorems of the theory contains all applications of the following axiom schemas:

- Reflexivity: For any variable  $x$ ,  $x = x$ .
- Leibniz's law or substitution: For any  $n \in \mathbb{N}_0$ , for any formula with any  $n$  free variables  $x_1, x_2, \dots, x_n$ , any  $i \in \mathbb{N} \wedge i \leq n$ , and any variable  $x$ ,

$$(x_i = x) \implies (P(x_1, x_2, \dots, x_i, \dots, x_n) \implies P(x_1, x_2, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n)).$$

### iii Equivalence of consistency and satisfiability

A theory over propositional or first-order language is consistent iff it is satisfiable.

The  $\Leftarrow$  direction is from the Gödel's completeness theorem and the  $\Rightarrow$  direction is the Henkin's model existence theorem.

### iv Elementary class or axiomatizable class

The elementary class or axiomatizable class of a first-order theory  $T$  is the class of all structures that satisfies  $T$ .

### v Downward Löwenheim–Skolem Theorem

If a first-order theory with signature  $\sigma$  has an infinite model  $\mathcal{M}$  of some cardinality  $\kappa \geq \aleph_0$ , then for every  $\mu$  such that  $\max(|\sigma|, \aleph_0) \leq \mu \leq \kappa$ , it has model of cardinality  $\mu$  that is an elementary substructures of  $\mathcal{M}$ .

### vi Upward Löwenheim–Skolem Theorem

If a first-order theory with signature  $\sigma$  has an infinite model  $\mathcal{M}$  of some cardinality  $\kappa \geq \aleph_0$ , then for every  $\mu \geq \kappa$ , it has model of cardinality  $\mu$  that is an elementary extension of  $\mathcal{M}$ .

## vii Compactness theorem

If a first-order theory  $T$  is such that all its finite subsets have models, then  $T$  has model.

## VII Gödel's incompleteness theorems

### i Gödel's first incompleteness theorem

Any recursively enumerable consistent deductive system that can interpret the Robinson arithmetic is syntactically incomplete.

### ii Gödel's second incompleteness theorem

Any recursively enumerable consistent deductive system that can interpret the Robinson arithmetic can not prove its consistency.

## VIII Abstract logic

### i Abstract logic

Let  $L$  be a formal language closed under logical connectives and with functionally complete set of logical connectives, and  $\mathcal{M}$  be all models (possibly designated to be allowed) for it. The ordered pair  $(L, \mathcal{M})$  is called an abstract logic.

### ii Isomorphism-invariant

An abstract logic,  $(L, \mathcal{M})$ , is isomorphism-invariant iff for every two models  $\mathcal{M}, \mathcal{N} \in \mathcal{M}$ , for every sentence  $\varphi \in L$ ,  $\varphi$  is isomorphism-invariant between  $\mathcal{M}$  and  $\mathcal{N}$ .

### iii Expressive power

All isomorphism-invariant abstract logics form a class  $\mathbb{L}$  with partial order  $\leq$ , called weaker than or less expressive than, equivalence relation  $=$ , and strict partial order  $<$ , called strictly weaker than or strictly less expressive than, of expressive power defined, for any  $(L_1, \mathcal{M}_1), (L_2, \mathcal{M}_2) \in \mathbb{L}$ , with the  $\mathcal{N}_1$  and  $\mathcal{N}_2$  defined such that all elements of  $\mathcal{N}_1 \cup \mathcal{N}_2$  have the same signature  $\sigma$  and domain  $N$ , that for any  $\mathcal{M} \in \mathcal{M}_1$ , there exists at least one  $\mathcal{N} \in \mathcal{N}_1$  such that there exists at least one isomorphism between  $\mathcal{M}$  and  $\mathcal{N}$ , and that for any  $\mathcal{M} \in \mathcal{M}_2$ , there exists at least one  $\mathcal{N} \in \mathcal{N}_2$  such that there exists at least one isomorphism between  $\mathcal{M}$  and  $\mathcal{N}$ , as:

$$(L_1, \mathcal{M}_1) \leq (L_2, \mathcal{M}_2) \iff \bigcup_{\mathcal{N} \in \mathcal{N}_1} \text{Def}(\mathcal{N}) \subseteq \bigcup_{\mathcal{N} \in \mathcal{N}_2} \text{Def}(\mathcal{N}),$$

$$(L_1, \mathcal{M}_1) = (L_2, \mathcal{M}_2) \iff (L_1, \mathcal{M}_1) \leq (L_2, \mathcal{M}_2) \wedge (L_2, \mathcal{M}_2) \leq (L_1, \mathcal{M}_1).$$

$$(L_1, \mathcal{M}_1) < (L_2, \mathcal{M}_2) \iff (L_1, \mathcal{M}_1) \leq (L_2, \mathcal{M}_2) \wedge \neg(L_2, \mathcal{M}_2) \leq (L_1, \mathcal{M}_1).$$

### iv Compactness property

For an abstract logic  $(L, \mathcal{M})$ , if for every theory  $T$  over its language, that all its finite subsets have model in  $\mathcal{M}$  implies that  $T$  has model in  $\mathcal{M}$ , then the abstract logic has compactness property.



Compactness property implies that for any theory has model in  $\mathcal{M}$ , there exists a sound finitary deductive system that derive it.

That for any theory of a model in  $\mathcal{M}$ , there exists a sound, finitary and semantically complete deductive system that derive it, implies compactness property.

#### **v Downward Löwenheim–Skolem property**

For an abstract logic  $(L, \mathcal{M})$  with signature  $\sigma$ , if for every theory  $T$  over  $L$  that has an infinite model  $\mathcal{M} \in \mathcal{M}$  of some cardinality  $\kappa \geq \aleph_0$ , for every  $\mu$  such that  $\max(|\sigma|, \aleph_0) \leq \mu \leq \kappa$ , it has model of cardinality  $\mu$  that is an elementary substructures of  $\mathcal{M} \in \mathcal{M}$ , then the abstract logic is called to have downward Löwenheim–Skolem property.

#### **vi Upward Löwenheim–Skolem Theorem**

For an abstract logic  $(L, \mathcal{M})$ , if for every theory  $T$  over  $L$  that has an infinite model  $\mathcal{M} \in \mathcal{M}$  of some cardinality  $\kappa \geq \aleph_0$ , then for every  $\mu \geq \kappa$ , it has model of cardinality  $\mu$  that is an elementary extension of  $\mathcal{M} \in \mathcal{M}$ .

#### **vii Lindström's theorem**

Any isomorphism-invariant abstract logic extending first-order logic with standard semantics that has compact property and downward Löwenheim–Skolem property must not has strictly more expressive power than first-order logic with standard semantics.